


```
from diffusers.utils import export_to_video
from datetime import datetime

# Same prompt as LTX and HunyuanVideo tests for direct comparison
prompt = "A woman with long brown hair smiling gently at the camera in a sunlit park, natural lighting, soft breeze moving her hair, cinematic quality"
negative_prompt = "blurry, low quality, distorted, unnatural, deformed"

generator = torch.Generator(device="cuda").manual_seed(42)

print(f"Prompt: {prompt}\n")
print(f"Generating video with Wan 2.2 T2V-A14B on A100...")

start_time = datetime.now()

video_frames = pipe(
    prompt=prompt,
    negative_prompt=negative_prompt,
    height=480,
    width=832, # Wan 2.2 uses 480p at 832x480 as its native resolution
    num_frames=81, # ~5 seconds at 16fps native
    num_inference_steps=40,
    guidance_scale=5.0, # Wan default guidance
    generator=generator,
).frames[0]

end_time = datetime.now()
duration = (end_time - start_time).total_seconds()

print(f"\nGeneration complete in {duration:.1f} seconds ({duration/60:.1f} minutes)")
print(f"Number of frames: {len(video_frames)}")
```

Prompt: A woman with long brown hair smiling gently at the camera in a sunlit park, natural lighting, soft breeze moving her hair, cinematic quality

Generating video with Wan 2.2 T2V-A14B on A100...

100% 40/40 [12:22<00:00, 18.54s/it]

Generation complete in 752.1 seconds (12.5 minutes)

Number of frames: 81

```
#Saving Wan output to Drive
from datetime import datetime
import json

timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
video_id = f"wan_001_{timestamp}"

video_path = f'{DRIVE_PROJECT}/videos/generated/wan/{video_id}.mp4'
metadata_path = f'{DRIVE_PROJECT}/metadata/{video_id}.json'

# Export video (Wan native fps is 16 but 24 for consistency with LTX and Hunyuan)
export_to_video(video_frames, video_path, fps=16)

# Save metadata in consistent format
metadata = {
    "video_id": video_id,
    "generator": "Wan 2.2 T2V-A14B",
    "generator_version": "Wan-AI/Wan2.2-T2V-A14B-Diffusers",
    "prompt": prompt,
    "negative_prompt": negative_prompt,
    "seed": 42,
    "width": 832,
    "height": 480,
    "num_frames": 81,
    "fps": 16,
    "num_inference_steps": 40,
    "guidance_scale": 5.0,
    "generation_time_seconds": duration,
    "generation_datetime": timestamp,
    "gpu_used": "A100",
    "optimizations": ["vae_tiling", "vae_slicing"]
}

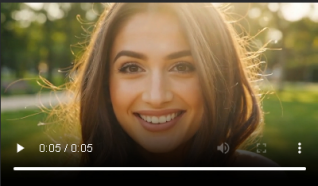
with open(metadata_path, 'w') as f:
    json.dump(metadata, f, indent=2)

print(f"Video saved: {video_path}")
print(f"Metadata saved: {metadata_path}")
```

Video saved: /content/drive/MyDrive/msc_deepfake_project/videos/generated/wan/wan_001_20260717_182819.mp4

Metadata saved: /content/drive/MyDrive/msc_deepfake_project/metadata/wan_001_20260717_182819.json

```
#Preview inline
from IPython.display import Video
Video(video_path, embed=True, width=400)
```



```
#Downloading to local machine
from google.colab import files
files.download(video_path)
```

...